

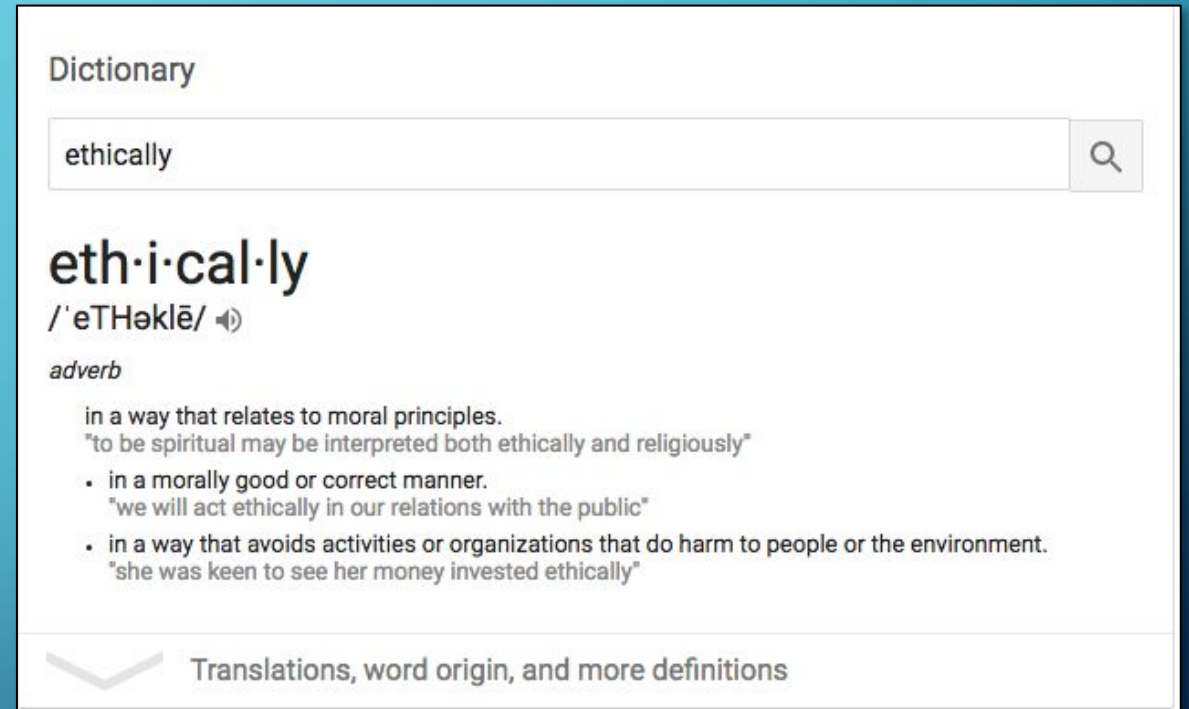
A decorative graphic on the left side of the slide, consisting of a network of light blue lines and small circles, resembling a circuit board or a stylized tree structure, set against a dark blue gradient background.

OPEN SOURCE INTELLIGENCE AND RECONNAISSANCE

JANUARY 4, 2019

ETHICS

- I am about to teach you how people gather targeting data on computer networks
- Everything I show you is publicly available and legal
- However it is crucial to always conduct this work ethically
 - Our commitment to ethical security assessment is key to our professional reputations
 - This is key to our ability to function in the greater security community



RECONNAISSANCE

- Also known as Open Source Intelligence (OSINT)
- In order to scan a system or network, you need to know...
 - Target Network Ranges OR Target System IP address
 - Other information found via OSINT/Recon that may be useful while scanning...
 - Systems of interest within the network
 - Insight into potential vulnerabilities

WHAT IS OSINT?

- OSINT = Open Source Intelligence
- Information that may be found on websites, forums, blogs, media, etc.
- The limitations of OSINT are not bound
- The ability to utilize information that is freely or near freely available
- Ability to utilize available resources of information to build intelligence

RECON-NG

- For our exercise today, we will use the open source tool, Recon-NG
 - Developed by Tim Tomes (LaNMaSteR53)
 - One of the many tools provided in Kali Linux

Recon-ng

Recon-ng is a full-featured Web Reconnaissance framework written in Python. Complete with independent modules, database interaction, built in convenience functions, interactive help, and command completion, Recon-ng provides a powerful environment in which open source web-based reconnaissance can be conducted quickly and thoroughly.

Recon-ng has a look and feel similar to the Metasploit Framework, reducing the learning curve for leveraging the framework. However, it is quite different. Recon-ng is not intended to compete with existing frameworks, as it is designed exclusively for web-based open source reconnaissance. If you want to exploit, use the Metasploit Framework. If you want to social engineer, use the Social-Engineer Toolkit. If you want to conduct reconnaissance, use Recon-ng! See the [Usage Guide](#) for more information.

Recon-ng is a completely modular framework and makes it easy for even the newest of Python developers to contribute. Each module is a subclass of the "module" class. The "module" class is a customized "cmd" interpreter equipped with built-in functionality that provides simple interfaces to common tasks such as standardizing output, interacting with the database, making web requests, and managing API keys. Therefore, all the hard work has been done. Building modules is simple and takes little more than a few minutes. See the [Development Guide](#) for more information.

Sponsors



<https://bitbucket.org/LaNMaSteR53/recon-ng>



CLASSROOM DEMO



Please follow along with the provided handout